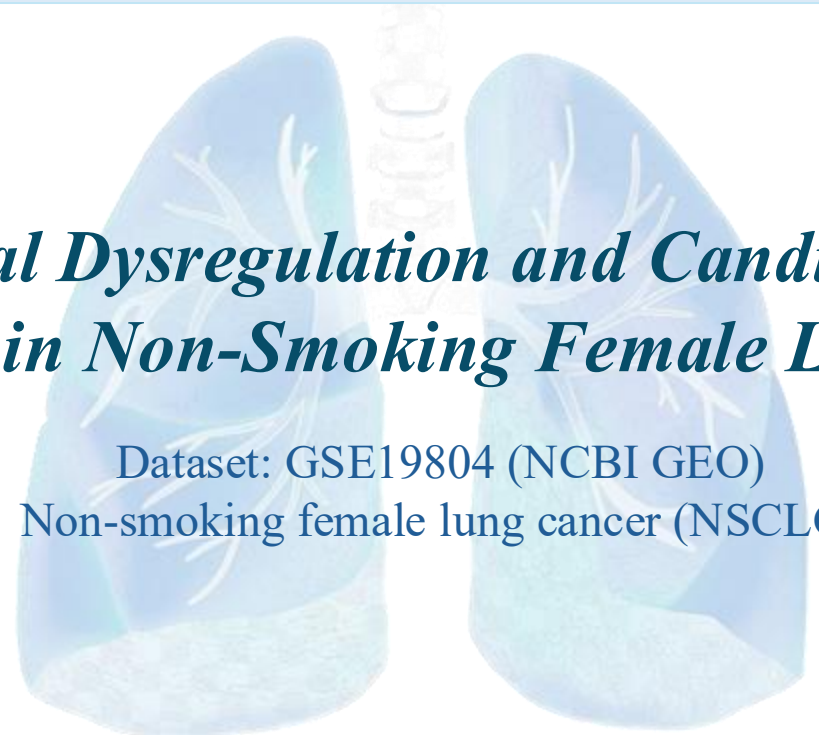




Transcriptional Dysregulation and Candidate Biomarker Discovery in Non-Smoking Female Lung Cancer

Dataset: GSE19804 (NCBI GEO)
Non-smoking female lung cancer (NSCLC)

Instructor: Diksha Marwaha
PhD Scholar, IIITD New Delhi

Presented by: Divyanshu Jha
St. Xavier's College, Ahmedabad

Dataset Summary

Accession	GSE19804
Organism	Homo sapiens
Platform	GPL570 - Affymetrix HG-U133 Plus 2.0
Design	Paired tumor / adjacent normal tissue
Samples	120 total (60 tumor + 60 matched normal)
Subtype	Non-smoking female lung cancer (Taiwan)

60 Vs 60

Perfectly balanced paired tumor/normal design

54,675

Probe sets profiled per sample (RMA-normalized)

Why this dataset?

Non-smoking female lung cancer is a clinically distinct subgroup with different molecular pathogenesis than smoking-associated disease, disproportionately affecting East Asian women. The paired design controls for inter-patient variability.

Differential Expression: Tumor Vs Normal

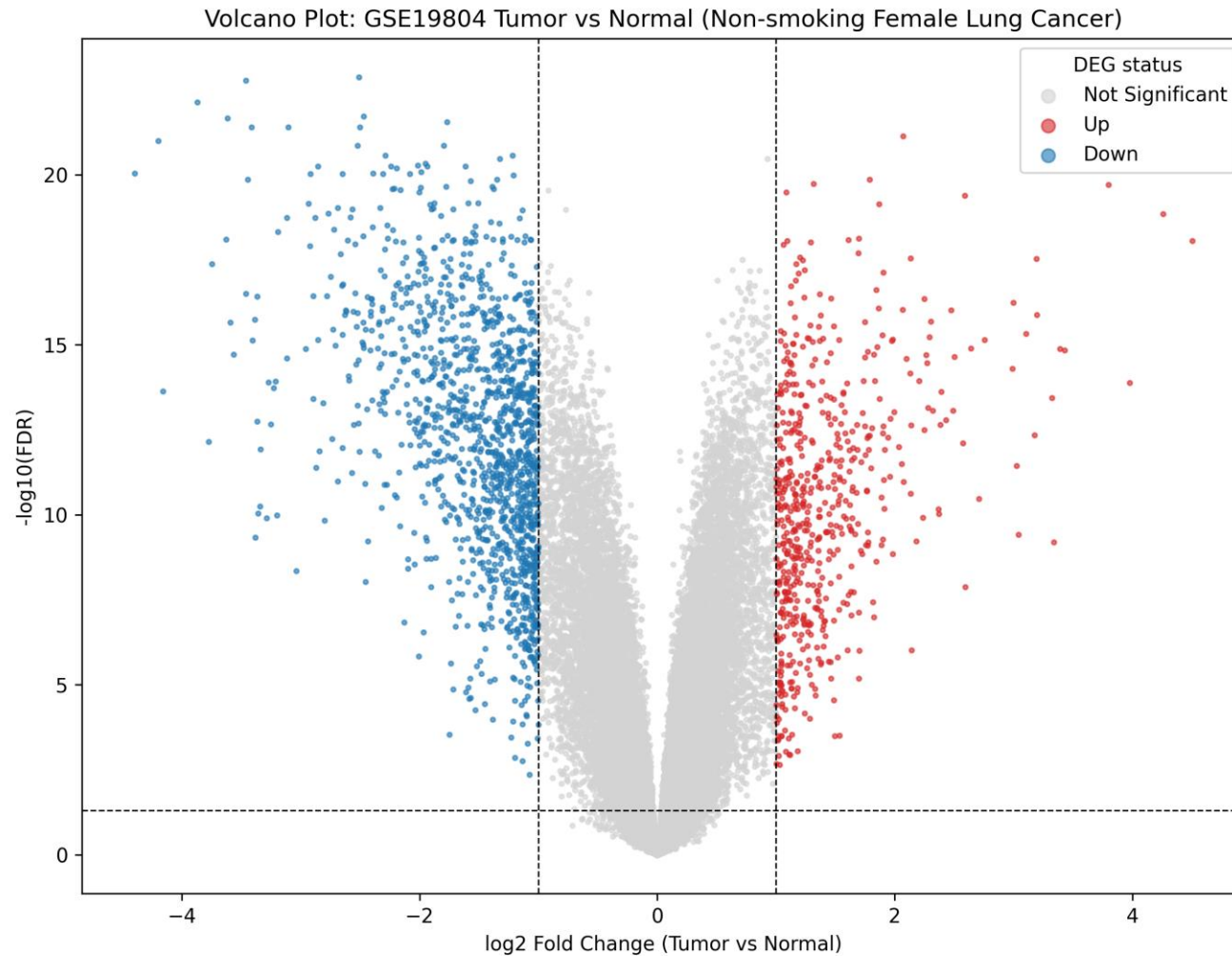


Figure: Volcano plot of differential expression. Red: significantly upregulated genes; blue: significantly downregulated genes; grey: not significant. Dashed lines indicate significance thresholds ($|\log_2\text{FC}| > 1$, $\text{FDR} < 0.05$).

1401

Significant DEG
 $\text{FDR} < 0.05$, $|\log_2\text{FC}| > 1$

453

Upregulated genes
Cell Cycle / Proliferation

948

Downregulated genes
Immune / vascular theme

Top hit: AGER ($\log_2\text{FC} = -3.46$, $\text{FDR} = 1.7 \times 10^{-23}$) - independently confirmed downregulated in lung cancer across multiple published studies.

Sample-Level Visualization

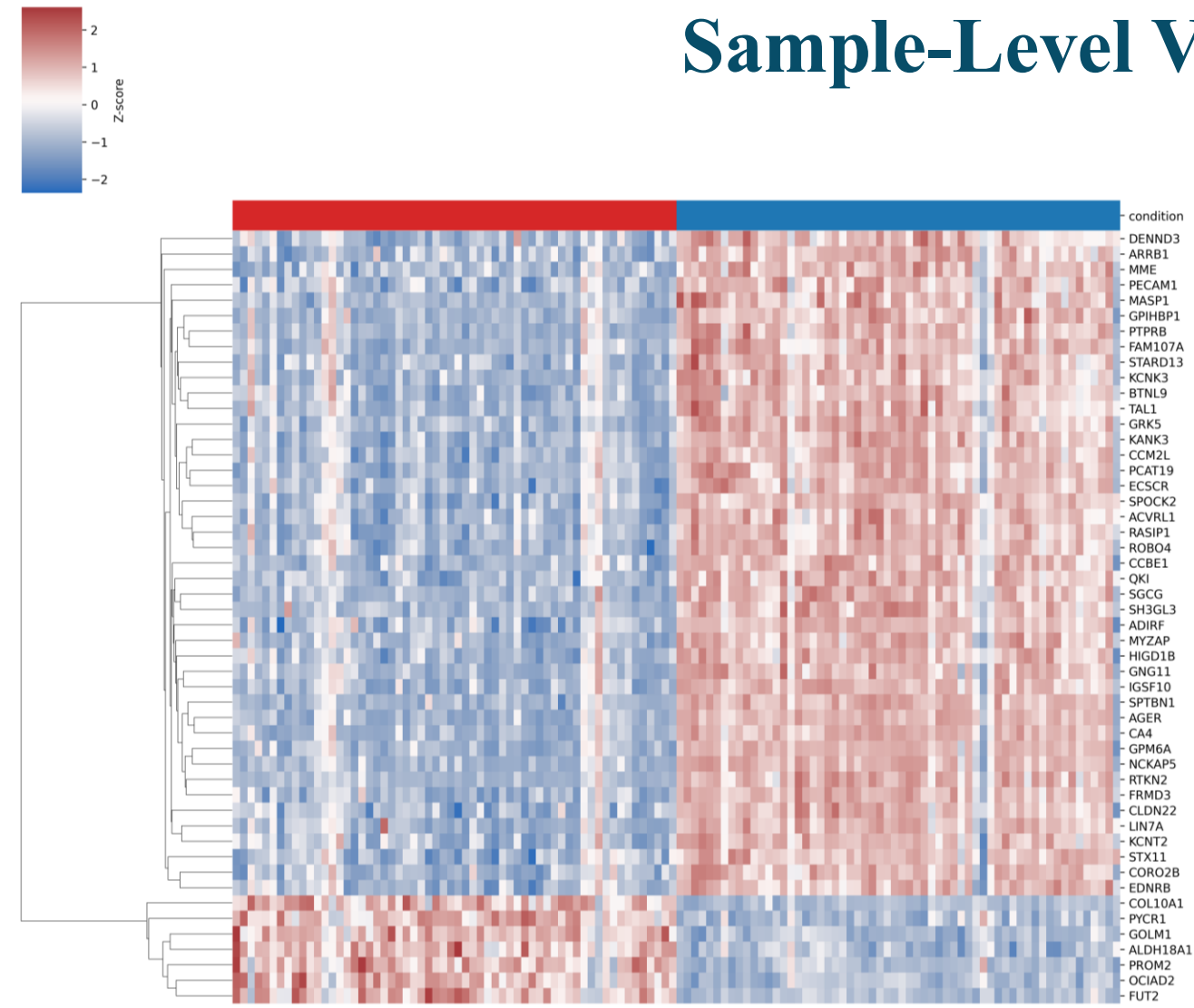


Figure: Top 50 DEGs cluster cleanly into Tumor (red) vs. Normal (blue) blocks

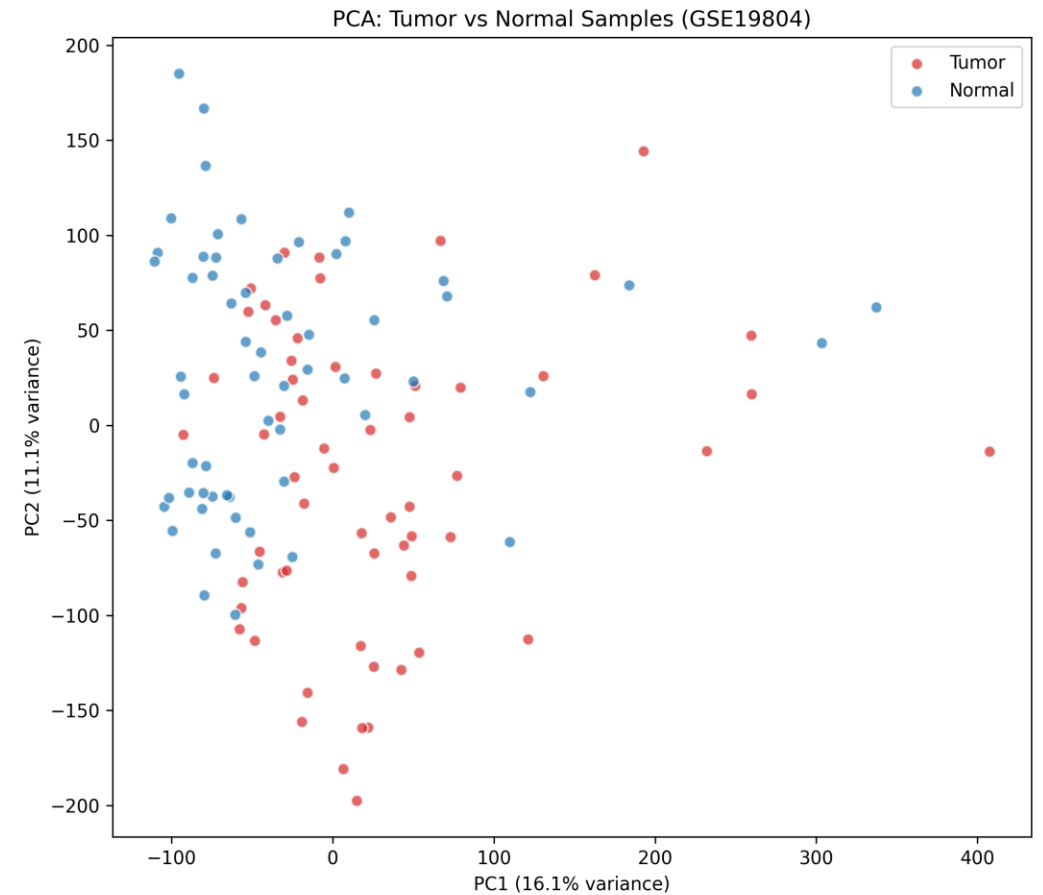


Figure: Global PCA (all genes) shows weak separation (~27% variance in PC1+PC2) - inter-patient variability dominates. This is why apaired statistical test was used for DEG analysis.

Functional Enrichment: GO & KEGG

UPREGULATED

Cell Cycle / Proliferation

- KEGG: Cell cycle (18/124, $p=8.5e-8$)
- KEGG: p53 signaling pathway
- GO: Mitotic spindle checkpoint signaling
- GO: Positive regulation of cell cycle process
- GO: Extracellular matrix organization

Interpretation: tumor tissue shows increased proliferative and mitotic checkpoint activity, consistent with uncontrolled cell division in cancer.

DOWNREGULATED

Immune Response / Vascular Signaling

- KEGG: Cytokine-cytokine receptor interaction
- KEGG: TNF signaling pathway
- KEGG: Chemokine signaling pathway
- GO: Regulation of angiogenesis
- GO: Inflammatory response

Interpretation: reduced immune surveillance and vascular signaling may partly reflect tissue composition differences in bulk RNA data.

PPI Network & Hub Genes

150

input genes (top 75 up + top 75 down)

40 / 36

connected nodes / high-confidence edges (STRING, score ≥ 700)

2

coherent biological clusters identified via degree centrality

Vascular / Endothelial Cluster

Top hub: CDH5 (degree = 7)

- PECAM1 (degree 3) - down, immune infiltration marker
- TEK (degree 3) - down, Tie2 angiogenesis receptor
- ROBO4 (degree 2) - down, favorable NSCLC prognostic marker

ECM / Collagen Cluster

2nd hub: COL1A1 (degree = 4)

- COL3A1 (degree 2) - up, cisplatin resistance marker
- COL10A1 (degree 2) - up, largest log2FC of all candidates
- Independently published as PPI hub in external LUAD study

Candidate Biomarkers & Validation

Selected using DEG strength + PPI hub status + pathway relevance
Not confirmed biomarkers - pending independent clinical validation

Gene	Direction	log2FC	Literature
AGER	Down	-3.46	Match
COL1A1	Up	+3.00	Match
PECAM1	Down	-1.88	Match
ROBO4	Down	-2.39	Match
ZWINT	Up	+1.77	Match
COL10A1	Up	+3.80	Match
COL3A1	Up	+1.98	Match
CDH5	Down	-2.20	Vascular caveat
TEK	Down	-2.60	Vascular caveat

Classifier Performance

AUC = 0.996

5-fold CV AUC = 0.993 +/- 0.010

Sensitivity 0.933 | Specificity 1.000 | F1 0.966

Patient-level split - no data leakage

7 / 9 genes direction-matched in literature

- ZWINT: meta-analysis of 2,847 LUAD patients
- COL1A1: external AUC 0.909 (with CEA)
- CDH5/TEK: vascular markers - bulk tissue composition likely explains direction difference

Clinical Insights, Limitations & Conclusion



Clinical Insight

A 9-gene panel spanning proliferation and vascular/ECM biology shows strong DEG, network, and literature support - a coherent candidate signature for further diagnostic evaluation in non-smoking female lung cancer.



Limitations

- Single dataset - generalizability not established
- Bulk tissue: cell-composition effects possible
- Classifier AUC reflects DEG strength, not independent validation

Future work: validate the 9-gene panel in an independent cohort (TCGA-LUAD), confirm protein-level expression via Human Protein Atlas, and apply single-cell deconvolution to resolve the CDH5/TEK direction discrepancy.

Conclusion

This project reproduced a complete tumor-vs-normal transcriptomics pipeline end to end from raw expression data to a literature-supported, multi-evidence candidate biomarker panel - while maintaining statistical rigor (paired testing, patient-level cross-validation) and interpretation of its own limitations.